The NASA observer's rating could be noted by the meteorological observer monitoring a binary light display on the NCAR Signal Conditioning Unit. This enabled the meteorological observer to compare the subjective analysis of ride quality from both the rear passenger cabin and cockpit of the aircraft.

As expected, the NASA observer was usually more critical of aircraft response to a turbulence encounter than the meteorological observer. It was also noted that when turbulence increased in intensity, the NASA observer generally responded to this condition with quick changes in the rating box scale to reflect the decrease in ride comfort. In contrast, the observer's reaction toward the improvement in ride quality was usually much slower when turbulence decreased in intensity. Furthermore, it was noticed that if the same degree of turbulence was encountered for a long period of time, the observer would increase his discomfort rating. This indicates that the duration of turbulence is also a factor in assessing passenger comfort. Of course, it is also a generally accepted fact that passengers respond more critically to turbulence in the tail section of a long swept wing airplane than passengers in front or the crew in the cockpit.

During the data collection program, rating box data were lost on three (3) flights and recorded intermittently on two (2) others. This was due to a faulty wire connection in the rating box and broken wires in the power supply connector. The observers on these flights took extensive notes to cover the loss in recorded rating box data.

Tail Section Accelerometers

NASA provided two Donner, Model 4310, single axis accelerometers that were installed in the overhead center ceiling area above the rear lavatories in the tail of the B-747. The accelerometers measured vertical and lateral motions of the tail section of the airplane.

The Donner linear accelerometer functions as a miniature servo system measuring input accelerations along its sensitive axis to an accuracy of 0.1%. The accelerometers operated on 28 volt DC power with the following range scales.

Vertical . . . $\pm$ 2.0 g full scale; calibrated $\pm$ 1.0 g
Lateral . . . $\pm$ 0.5 g full scale; calibrated $\pm$ 0.5 g